

Compositional Realization, Hamiltonian Reconstruction, and Empirical Limits: a Synthesis

What the three-paper series proves jointly, where its papers pull against each other, and
what its formal appendix does not certify

PRINCIPIA PHYSICA pilot — synthesis node

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Abstract

This paper synthesises three companion drafts of the PRINCIPIA PHYSICA pilot: *Compositional Systems in Finite-Dimensional Classical Mechanics* (hereafter CS), *Hamiltonian Reconstruction as a Physical Theory Object* (HR), and *Empirical Realization, Equivalence, and Limits* (EL). A synthesis is a typed merge, not an amplification: no claim acquires a stronger epistemic type by appearing in more than one paper, and consensus among the drafts is never treated as proof. We consolidate the definitions into one three-layer object (compositional structure, generator, registered interpretation), give a dependency ledger keyed to the companion papers' claim identifiers, and prove four statements that are visible only across papers: the additive constant of a generator is invisible to every registered experiment (Proposition 4.1); factor-level empirical adequacy plus the monoidal product does not entail composite adequacy, with an explicit tolerance witness (New result 4.5); bare isomorphism and ε -empirical equivalence are incomparable in the precise sense supplied by HR's two counterexamples, with one residual gap recorded (Proposition 4.7); and no arrow class considered in the series simultaneously preserves Hamiltonian structure, contains coarse-graining, and composes totally (Proposition 4.8). We preserve, rather than resolve, three disagreements between the drafts and four failed constructions. A machine-checked Lean file and a type-checked Haskell file accompany the paper; Section 6 states exactly which claims they do and do not certify, and the answer is: none of the smooth theorems.

1 What this paper is, and what a synthesis may not do

The pilot's founding thesis is that physics is the study of empirically realized compositional mathematical structures. All three companion papers carry that sentence as a *conjecture* and use it as a premise nowhere. This paper does the same.

Three constraints govern what follows. First, every displayed statement carries exactly one epistemic type from the pilot contract: Definition, Axiom, Assumption, Conjecture, Lemma, Proposition, Theorem, Corollary, Interpretation, Empirical statement, Established physical result, or New result claimed by this work. Second, a claim imported from a companion paper keeps the type it had there; agreement between two drafts is not evidence and does not promote a conjecture to a theorem. Third, where the drafts conflict, or where a construction they attempt fails, the conflict and the failure are recorded in Section 5 rather than smoothed away.

Scope is inherited from the pilot charter: smooth, finite-dimensional, real classical systems. Field theory, infinite-dimensional analysis, general relativity, and quantum theory appear only as boundary markers, and each such appearance below is an Established physical result cited to its source or an explicit statement that the present type stops there. No new experimental claim is made anywhere in this paper.

2 The consolidated typed object

The three drafts define overlapping objects with different names. We fix one vocabulary and record the correspondence.

Definition 2.1 (Three-layer object). *A three-layer object is a triple $(\mathcal{C}, \mathsf{T}, \mathcal{I})$ in which*

- (L1) $\mathcal{C}$ is a symmetric monoidal category of systems and processes (CS Definitions of system, process, and composite; HR Definition of morphism and Cartesian product);
- (L2) $\mathsf{T} = (M, \omega, \mathcal{A}(M), H, \Phi)$ is a bare Hamiltonian object in $\mathcal{C}$: a symplectic manifold, its observable algebra, a generator, and the maximal local flow of the field X_H determined by $\iota_{X_H}\omega = dH$ (HR Axioms on kinematic type, symplectic structure, observable type, generator, dynamical law);
- (L3) $\mathcal{I} = (\mathcal{C}_{\text{ctx}}, K^P, K^Q, \varepsilon, D)$ is a registered interpretation: experimental contexts, preparation and measurement kernels, a declared tolerance, and an intended domain (EL Definition of empirical interpretation; HR $\mathcal{I} = (P, \mu, \mathcal{U}, \varepsilon, T)$ is the deterministic special case, augmented by units).

Write $U(\mathcal{C}, \mathsf{T}, \mathcal{I}) = (\mathcal{C}, \mathsf{T})$ for the forgetting of layer (L3).

Definition 2.2 (Predicted law, realization, discrepancy). *For a context $c \in D$ with control u_c and duration t_c , the predicted outcome law is*

$$\Pr_{\mathsf{T}, \mathcal{I}}(A \mid c) = \int_M K_c^Q(A \mid \Phi_{t_c}^{u_c}(x)) K_c^P(dx). \quad (1)$$

A physical target S realizes the object on D to tolerance ε when $d_{\text{TV}}(P_S^c, \Pr_{\mathsf{T}, \mathcal{I}}(\cdot \mid c)) \leq \varepsilon(c)$ for every registered $c \in D$. For two objects sharing the registered outcome spaces, $\Delta_D = \sup_{c \in D} d_{\text{TV}}(P_1^c, P_2^c)$, and they are η -observationally equivalent on D when $\Delta_D \leq \eta$.

Definition 2.2 is EL’s, transcribed. Its role here is that layers (L1) and (L2) never enter an empirical claim except through (1); that is the single channel through which mathematics reaches data in this series, and every result below is a statement about what does or does not pass through it.

Axiom 2.3 (Typed equality). *States, processes, observables, interpretations, and empirical records are distinct types. An equality is admitted only between terms of one type; a map relating two types is part of the data and is never suppressed.*

Axiom 2.3 is CS’s, and it is what makes Proposition 4.7 statable rather than paradoxical.

3 Dependency ledger: what the series proves

Table 1 lists the imported claims used below, with the identifier under which each appears in its companion paper’s metadata. The ledger is exhaustive for the results this synthesis *uses*; it is not a census of the companion papers, and Section 5 records that the companion metadata is itself incomplete.

4 Results visible only across the three papers

Each result in this section uses inputs from at least two companion papers. Each is elementary given those inputs; the claim is to the explicit cross-paper statement, and where the argument is a quantified restatement of a companion result we say so in the same breath.

Identifier	Type	Content used here
CS-D1	definition	Composite system $(M \times N, \pi_M^* \omega + \pi_N^* \eta)$.
CS-P1	proposition	Symplectic systems and smooth maps form a symmetric monoidal category $\mathbf{Ham}_{\text{kin}}$.
CS-T1	theorem	An additively separable generator has componentwise field and product flow.
CS-N1	new result	Converse: on a connected product the field splits iff the generator is separable up to a constant.
CS-I1	interpretation	Product composition represents physical independence only relative to operational separability.
HR ax-symplectic, ax-dynamical-law	axiom	$d\omega = 0$, ω nondegenerate; Φ is the maximal local flow of X_H .
HR lem-hamiltonian-field	lemma	X_H exists and is unique.
HR thm-exact-sequence	theorem	$0 \rightarrow H_{\text{dR}}^0 \rightarrow \mathcal{A}(M) \rightarrow \mathfrak{X}_{\text{Symp}} \rightarrow H_{\text{dR}}^1 \rightarrow 0$ is exact.
HR cor-underdetermination	corollary	On connected M the flow determines H up to one additive constant.
HR lem-symplectic-immersion	lemma	$\varphi^* \omega' = \omega$ forces $d\varphi$ injective.
HR new-no-coarse-graining	new result	No morphism decreases dimension.
HR new-interaction-outside-tensor	new result	On a split product, $\Phi^H = \Phi^1 \times \Phi^2$ iff the coupling is locally constant.
HR prop-equivalent-not-isomorphic	proposition	Grönwall witness: ε -equivalent, non-isomorphic.
HR prop-isomorphic-not-physically-equivalent	proposition	Unit witness: isomorphic bare objects, distinct empirical models.
HR new-damped-no-symplectic-form	new result	A globally asymptotically stable planar field is Hamiltonian for no symplectic form.
HR est-weinstein-not-category	established	Canonical relations do not compose totally.
HR est-groenewold-van-hove	established	No structure-preserving quantization functor on $\mathbb{R}^{2n}$.
HR est-darboux, est-gromov-nonsqueezing	established	Local triviality; symplectic rigidity.
EL C2	theorem	Intertwining of preparation, dynamics, and measurement preserves every registered law.
EL C3	new result	Uniform error transfer: $\sup_c d_{\text{TV}}(P_S, P_0) \leq \sup_c e_\lambda + r(\lambda)$.
EL C4	proposition	Structural equivalence alone does not give physical equivalence.
EL restriction lemma	lemma	$D' \subseteq D$ implies $\Delta_{D'} \leq \Delta_D$.

Table 1: Imported claims, with their companion-paper identifiers. Types are those assigned in the source paper and are not upgraded here.

4.1 The additive constant is empirically idle

Proposition 4.1 (Constant invisibility). *Let $(\mathcal{C}, \mathbb{T}, \mathcal{I})$ be a three-layer object with connected M , and let $\mathbb{T}_c$ be obtained by replacing H with $H + c$ for a constant $c \in \mathbb{R}$. Then for every registered context set D and every choice of kernels in $\mathcal{I}$,*

$$\Pr_{\mathbb{T}, \mathcal{I}}(\cdot \mid c') = \Pr_{\mathbb{T}_c, \mathcal{I}}(\cdot \mid c') \quad \text{for all } c' \in D, \quad \text{hence} \quad \Delta_D(\mathbb{T}, \mathbb{T}_c) = 0.$$

Conversely, if $\mathbb{T}' = (M, \omega, \mathcal{A}(M), H', \Phi')$ has $\Phi' = \Phi$ on a neighbourhood of $t = 0$, then $H' = H + c$ for some constant c .

Proof. By HR lem-hamiltonian-field, X_{H+c} is the unique field with $\iota_{X_{H+c}}\omega = d(H+c) = dH$, so $X_{H+c} = X_H$, and two vector fields that are equal have the same maximal local flow: $\Phi^{H+c} = \Phi^H$. Both sides of (1) then integrate the same kernel against the same pushforward, so the predicted laws agree context by context, and the supremum of a family of zeros is zero. For the converse, equality of the flows near $t = 0$ gives equality of their generators' fields by differentiation at $t = 0$, and HR cor-underdetermination on connected M gives $H' - H$ locally constant, hence constant. $\square$

Corollary 4.2 (The observational map is not injective). *The assignment sending a bare object to its observational class on D is not injective on generators: it identifies H with $H + c$ for every c . No enlargement of D within Definition 2.2 separates them.*

Proof. Proposition 4.1 gives $\Delta_D = 0$ for every D , and the EL restriction lemma states only that shrinking D cannot increase Δ ; enlarging D leaves the computation of (1) unchanged because the flows are equal. $\square$

Interpretation 4.3 (Zero of energy). *Within this scope an absolute zero of energy is not an empirically realized element of the object: it is a coordinate on the space of generators that (1) annihilates. HR records the same reading at int-energy-zero. That gravitation changes the situation is a boundary fact, outside the present type, and is not used.*

4.2 Factor adequacy does not compose

CS supplies the counterexample that interaction defeats product dynamics; HR sharpens it to an exact characterisation of the range of $\otimes$; EL supplies the tolerance language in which “adequate” is defined. Only the combination states what an experimenter would need.

Assumption 4.4 (Registered isolated adequacy). *Let the target be a pair of laboratory oscillators that can be operated singly. We assume: (i) for the isolated contexts D_1 (one subsystem prepared and read, the other absent), the uncoupled objects $\mathbb{T}_i = (T^*\mathbb{R}, dq_i \wedge dp_i, H_i)$ with $H_i = (p_i^2 + q_i^2)/2$ realize the target exactly, with noiseless readout $\mu(z) = q_1$ in metres; (ii) the joint contexts D_{12} prepare both subsystems together. Clause (i) is an empirical premise, not a theorem, and it is chosen exact only to isolate the compositional failure from measurement error.*

New result claimed by this work 4.5 (Composite adequacy is not inherited). *Adopt Assumption 4.4, and let the joint target be governed by*

$$H^\kappa = \frac{1}{2}(p_1^2 + p_2^2) + \frac{1}{2}(q_1^2 + q_2^2) + \frac{\kappa}{2}(q_1 - q_2)^2, \quad \kappa > 0,$$

on $(T^\mathbb{R})^{\otimes 2}$ with the split form. Let $\mathbb{T}_1 \otimes \mathbb{T}_2$ be the monoidal composite of the two factor objects. Then for the joint context that prepares $z_0 = (q_1, q_2, p_1, p_2) = (1, -1, 0, 0)$ and reads $\mu = q_1$ at time $t_\kappa := \pi/\sqrt{1+2\kappa}$,*

$$|\mu(\Phi_{t_\kappa}^\kappa z_0) - \mu((\Phi^1 \times \Phi^2)_{t_\kappa} z_0)| = 1 + \cos\left(\frac{\pi}{\sqrt{1+2\kappa}}\right) =: \delta(\kappa) > 0.$$

The coordinate separation is $\delta(\kappa)$, whereas the total-variation discrepancy of the noiseless outcome laws is exactly 1. Consequently, for every total-variation tolerance $\varepsilon < 1$ and every window $T \geq t_\kappa$, the factor objects realize the target on D_1 while $\mathsf{T}_1 \otimes \mathsf{T}_2$ fails to realize it on D_{12} . Moreover $\delta(\kappa) \rightarrow 0$ as $\kappa \downarrow 0$ and $\delta(\kappa) \rightarrow 2$ as $\kappa \rightarrow \infty$.

Proof. Set $u = (q_1 + q_2)/\sqrt{2}$, $v = (q_1 - q_2)/\sqrt{2}$ and correspondingly for the momenta; this is a linear symplectic change of coordinates, so the split form is preserved and $H^\kappa = \frac{1}{2}(p_u^2 + p_v^2) + \frac{1}{2}u^2 + \frac{1}{2}(1 + 2\kappa)v^2$. The two normal modes are uncoupled harmonic oscillators with frequencies $\omega_+ = 1$ and $\omega_- = \sqrt{1 + 2\kappa}$. The initial datum z_0 has $u = 0$, $p_u = 0$, $v = \sqrt{2}$, $p_v = 0$, so $u(t) \equiv 0$ and $v(t) = \sqrt{2}\cos(\omega_-t)$, whence $q_1(t) = \cos(\omega_-t)$.

By CS-T1 (equivalently HR prop-product-well-typed), the composite $\mathsf{T}_1 \otimes \mathsf{T}_2$ has generator $H_1 \circ \pi_1 + H_2 \circ \pi_2$ and flow $\Phi_t^1 \times \Phi_t^2$, so its prediction for the same datum is $q_1(t) = \cos t$. At $t_\kappa = \pi/\omega_-$ the target value is $\cos \pi = -1$ and the composite's prediction is $\cos(\pi/\omega_-)$. Since $\kappa > 0$ gives $\omega_- > 1$, we have $0 < \pi/\omega_- < \pi$ and therefore $\cos(\pi/\omega_-) > -1$; the absolute difference is $1 + \cos(\pi/\omega_-) = \delta(\kappa) > 0$. Thus the predicted point masses are distinct. Under Definition 2.2, their total-variation distance is exactly 1, so realization fails precisely for tolerances $\varepsilon < 1$. The limits stated for the separate coordinate diagnostic follow by continuity of $\kappa \mapsto 1 + \cos(\pi/\sqrt{1 + 2\kappa})$ at the endpoints. $\square$

Interpretation 4.6 (Reading of New result 4.5). *The mathematics here is a quantified restatement of HR new-interaction-outside-tensor: because $\kappa \neq 0$ makes the coupling non-constant, the target's flow is outside the range of $\otimes$, and the only contribution of this paper is the explicit coordinate diagnostic $\delta(\kappa)$ and its separation from the total-variation discrepancy. Under noiseless readout the latter jumps from 0 to 1 for every nonzero coupling; it does not inherit the smooth small-coupling behavior of $\delta(\kappa)$. Two readings must be resisted. It does not show that composition fails in general. EL's compositional error bound remains a conditional positive result under its independence assumption, but this noiseless TV example supplies no small-error regime. It does show that the assumption is load-bearing, and that reporting factor adequacy without an interface test reports nothing about the composite.*

4.3 Isomorphism and empirical equivalence are incomparable

Proposition 4.7 (Incomparability, with a recorded residual). *Consider bare isomorphism $\cong$ (HR Definition of morphism) and ε -empirical equivalence $\sim_\varepsilon$ on (P, T) (HR Definition of ε -empirical equivalence). Then:*

- (i) $\sim_\varepsilon$ does not imply $\cong$. *Witness: the harmonic object and its quartic perturbation $H_\lambda = H_0 + \lambda q^4$ for λ small (HR prop-equivalent-not-isomorphic).*
- (ii) $\cong$ does not imply identity of empirical models. *Witness: the mass-spring and LC objects with $m = L$, $k = 1/C$, whose bare objects are equal and whose empirical models are records in metres and in coulombs respectively (HR prop-isomorphic-not-physically-equivalent).*

The two relations are therefore incomparable in this precise sense. *Residual: (ii) exhibits objects whose empirical models are not comparable without a translation of units, not two objects with a common registered outcome space and different laws. Whether $\cong$ can fail to imply $\sim_\varepsilon$ after a fixed unit translation and a common registered outcome space is not settled by the series, and is recorded as Open problem 4.10.*

Proof. (i) HR proves a λ -independent Grönwall bound $\sup_{z \in P} \sup_{t \in [0, T]} \|\Phi_t^\lambda z - \Phi_t^0 z\| \leq C\lambda$ and, separately, that the period function of H_λ is non-constant while that of H_0 is constant, which blocks any isomorphism because isomorphisms intertwine flows and hence preserve periods. Choosing $\lambda < \varepsilon/C$ gives $\sim_\varepsilon$ without $\cong$. (ii) HR exhibits the identity of $\mathbb{R}^2$ as an isomorphism of bare objects and observes that U deletes the unit assignment $\mathcal{U}$, so no bare isomorphism

induces a map of empirical models. By Axiom 2.3 the records are of different types, so the equality of empirical models is not merely false but ill-typed. Combining the two witnesses gives incomparability. $\square$

Proposition 4.8 (No arrow class in the series does all three jobs). *Let $\mathcal{A}$ be one of the three arrow classes considered in the series: (a) all smooth maps (CS’s Ham_{kin}); (b) maps with $\varphi^*\omega' = \omega$ and $H = H' \circ \varphi + c$ (HR’s morphisms); (c) canonical relations. Then no one of them has all three properties: preservation of Hamiltonian structure, containment of a dimension-decreasing (coarse-graining) arrow, and totally defined composition.*

Proof. Class (a) contains the dilation $f(q, p) = (2q, p)$, for which $f^*(dq \wedge dp) = 2dq \wedge dp \neq dq \wedge dp$ (CS counterexample), so it fails preservation. Class (b) satisfies preservation by definition, and by HR lem-symplectic-immersion every arrow is an immersion, so $\dim M \leq \dim M'$ and HR new-no-coarse-graining excludes dimension decrease; it fails containment. Class (c) contains reductions but, by HR est-weinstein-not-category, the set-theoretic composite of two canonical relations need not be a submanifold and need not be Lagrangian when it is, so composition is partial; it fails total composability. $\square$

Open problem 4.9. *Proposition 4.8 is a statement about the three classes the series considers, not a proof that no class succeeds. Whether some class of arrows — spans, decorated cospans, or the loose arrows of a double category — achieves all three properties simultaneously for finite-dimensional symplectic objects is open, and is the content of HR conj-interconnection-completeness.*

Open problem 4.10. *Do there exist two theory objects with isomorphic bare objects, a common registered outcome space, a fixed unit translation, and $\Delta_D > \varepsilon$? A positive answer would strengthen Proposition 4.7(ii) from a typing obstruction to a measured discrepancy; a negative answer would show that after unit translation isomorphism does imply empirical equivalence within this scope.*

5 Preserved disagreements and failed constructions

A synthesis that reported only the compatible parts would misrepresent the series. Four failures and three tensions are recorded. None is resolved here.

Tension 1: the compositional verdict. CS concludes with a layered positive result — symmetric monoidal kinematics proved, factorized dynamics proved for separable generators, physical compositionality left to empirical assumption. HR concludes that the compositional reading is *incomplete in a provable way*: $\otimes$ cannot express interaction, the richer composition is partial, and the category has no coarse-graining arrows. These are logically compatible — CS proves that a monoidal structure exists, HR computes the range of that structure — but they support different summaries, and the difference matters for the thesis. The consolidated position taken here: the existence claim is proved (CS-P1), the sufficiency claim is refuted for $\otimes$ alone (HR new-interaction-outside-tensor), and New result 4.5 shows the refutation survives translation into tolerance language.

Tension 2: which arrows are the physics. CS admits all smooth maps as processes, so that preparation and readout maps exist, and then must add a counterexample showing such arrows preserve neither the symplectic form nor Poisson brackets. HR admits only form-preserving maps and then must record that coarse-graining leaves the category. Proposition 4.8 states the cost of each choice. The series does not select between them, and this synthesis does not either.

Tension 3: metadata integrity of the series. The round-four series review returned verdict MAJOR with five blocking findings, all of them about the correspondence between each paper’s LaTeX and its machine-readable claim list: CS exported 7 claims against 27 epistemic environments in the source; HR exported 44 against 53; EL exported 5, with dependency fields containing free text (for example “Definitions of empirical interpretation and indexed observational equivalence”) instead of claim identifiers; and CS-T1 depends on an identifier CS-L1 that its own metadata omits. This synthesis repairs the ledger only for the claims it uses (Table 1, where every dependency is an identifier or a named established result). It does not repair the companion papers’ own metadata, and the review’s finding stands against them.

Failed construction 1: the symplectic category. Objects symplectic manifolds, arrows canonical relations, is the natural repair for the expressive deficit of $\otimes$. It is not a category: the composite of two canonical relations need not be a submanifold, and need not be Lagrangian when it is (HR est-weinstein-not-category). Transversality or clean-intersection hypotheses restore composition only partially.

Failed construction 2: a quantization functor. No linear, unit-preserving map from the Poisson algebra of polynomials on $\mathbb{R}^{2n}$ to self-adjoint operators sends $\{f, g\}$ to $\frac{1}{i\hbar}[\hat{f}, \hat{g}]$ and acts irreducibly on linear polynomials (HR est-groenewold-van-hove). The obstruction is a theorem, so the typed object does not extend functorially past its boundary. Quantum theory is a boundary marker for this series and nothing is claimed inside it.

Failed construction 3: dissipation inside the type. A globally asymptotically stable planar field is Hamiltonian for *no* symplectic form (HR new-damped-no-symplectic-form), so no change of geometry rescues damped systems within the type on that domain. Time-dependent or restricted-domain presentations exist and are standard; they are different objects.

Failed construction 4: formalizing the smooth theorems. The intended formal appendix for this series was a machine-checked verification of its central results. That was not achieved and is not claimed. What was achieved is described next.

6 The formal companions, and what they do not certify

Two files accompany this paper: `formal/lean/Proofs.lean` and `formal/haskell/Core.hs`.

Empirical statement 6.1 (Verification record). *The following are facts about this build environment, reported as run, not as claims about the mathematics. `Proofs.lean` was checked by Lean version 4.33.0 with no imports beyond the core prelude, no `Mathlib`, and no occurrence of `sorry`; `#print axioms` on each of its nineteen theorems reports dependence only on `propext` and `Quot.sound` (four of them on no axiom at all), and no theorem depends on `sorryAx` or on the axiom of choice. `Core.hs` was compiled by GHC 9.14.1 on `aarch64-apple-darwin` under `-Wall -Wextra -Werror -fforce-recomp`, which reported no diagnostic and linked an executable; running that executable exits 0 and reports all 24 self-checks as `PASS` and none as `FAIL`. A compilation and an exit status are facts about this host and this toolchain, and about nothing else: neither is evidence for any statement of the series, and Interpretations 6.2 and 6.3 bound what the two files can be read as certifying.*

Interpretation 6.2 (What the Lean file certifies). *It certifies six discrete surrogates and nothing else. In the surrogate a manifold is a bare type, a symplectic form is absent, a vector field is a step function, and the first de Rham cohomology of the circle is replaced by the cycle sum of a periodic integer sequence. The checked statements are: the run of a monoidal product factors*

into the runs of its factors; every product is non-interacting and an explicitly coupled system is therefore not a product; a coupling is additively separable exactly when its mixed second difference vanishes; a difference operator determines its potential up to an additive constant; a periodic form admits a global potential exactly when its cycle sum vanishes, with the constant form on an n -cycle as an explicit obstruction; and a property is identifiable exactly when it factors through the observational quotient. These are surrogates of CS-P1, HR new-interaction-outside-tensor, CS-N1, HR cor-underdetermination, HR thm-exact-sequence with its torus non-example, and EL C2, respectively. A surrogate that checks does not certify the smooth statement. Nothing in the file mentions a manifold, a 2-form, a flow, a probability kernel, or a tolerance, so no theorem of the series is verified by it.

Interpretation 6.3 (What the Haskell file certifies). *Less.* Its type declarations render Definition 2.1 and Definition 2.2 as types, which fixes arities and dependencies and makes the deletion performed by U visible: units are a field of the interpretation record and are unreachable from the bare object. Its checks are of two kinds, marked in the source. Those marked **DEC** are exact on the finite data supplied — separability on a grid, cycle sums, observational classes of a three-element model class — and their correctness criteria are the statements proved in the Lean file. Those marked **NUM** are floating-point illustrations with untracked discretisation and rounding error: RK4 trajectories exhibiting the Grönwall-type separation of Proposition 4.7(i), the decay of the damped field, and total-variation arithmetic. A **NUM** check that passes is an illustration and is not evidence.

Open problem 6.4. Formalize, in a proof assistant with a differential-geometry library, at least HR lem-hamiltonian-field and HR cor-underdetermination, and then Proposition 4.1 on top of them. The surrogate development here indicates only that the combinatorial skeleton is consistent; the smooth content — nondegeneracy, Cartan’s formula, the Poincaré lemma — is exactly what the surrogate omits.

7 Falsification conditions for the synthesised object

Definition 7.1 (Refutation of the type on a domain). *The three-layer type is refuted for a domain D of physical systems when there are a system in D , a registered interpretation, and recorded data $\mathcal{D}$ such that no object satisfying the HR axioms on any finite-dimensional M , with any registered interpretation, satisfies Definition 2.2’s realization criterion for $\mathcal{D}$.*

Four registered targets, stated as targets and not as claims, and each with the companion result that makes it discriminating:

- (F1) *Bracket failure*: measured response functions of a closed finite-dimensional system whose induced bracket violates the Jacobi identity beyond tolerance. This refutes $d\omega = 0$ for that system (HR lem-poisson-identities).
- (F2) *Global generator failure*: measured evolution preserving ω with a measurably nonzero class $[\iota_X\omega] \in H_{\text{dR}}^1(M)$. This refutes the dynamical-law axiom while leaving the kinematic axioms intact (HR cor-underdetermination; the Lean file’s cycle-sum obstruction is the discrete shadow of this test).
- (F3) *Irreducible dissipation*: a closed finite-dimensional system with a globally asymptotically stable equilibrium. By HR new-damped-no-symplectic-form this refutes the dynamical law under *every* symplectic form on its state space.
- (F4) *Interface failure*: two subsystems each realized to tolerance ε on isolated contexts whose joint records exceed ε on joint contexts by more than the EL compositional bound permits. By New result 4.5 this refutes the independence assumption rather than the type, which is what makes it a diagnostic rather than a global test.

Empirical statement 7.2 (Evidential status). *None of (F1)–(F4) is reported as observed here; no new measurement is presented in this paper. The series’ only empirical anchor is the cited adequacy of laboratory mass–spring and LC objects within stated regimes (HR emp-laboratory-anchor), and its only cited refutation of a specific object is the perihelion advance of Mercury against the two-body Newtonian Hamiltonian [13], which refutes that object and not the type.*

8 Open conjectures carried forward

Conjecture 8.1 (Interconnection completeness; HR). *There is a symmetric monoidal double category whose loose arrows are partially-defined canonical relations and whose tight arrows are symplectomorphisms, in which every object $(M_1 \times M_2, \omega_1 \oplus \omega_2, H_1 \oplus H_2 + V)$ arises as a composite of the factors and one arrow determined by V .*

Conjecture 8.2 (Compositional empirical adequacy; EL). *For a useful class of finite-dimensional classical laboratory systems, interpretations and error budgets can be assigned compositionally so that subsystem adequacy plus verified interface assumptions entails system-level adequacy with nonvacuous bounds.*

Conjecture 8.3 (Joint form; this paper). *Conjectures 8.1 and 8.2 are the mathematical and the empirical halves of one statement: that the interconnection arrow determined by a coupling V can be assigned an error budget depending only on V and on the interface, in such a way that New result 4.5’s $\delta(\kappa)$ is recovered as the special case of a quadratic coupling. We do not assert this; it is offered as the shape a positive result would have to take, and it inherits every obstruction recorded in Section 5.*

Conjecture 8.4 (Empirical bite of symplectic rigidity; HR). *There exist a finite-dimensional classical laboratory system, an interpretation, and two candidate objects agreeing on all volume-theoretic predictions within tolerance whose readout predictions differ by more than the tolerance solely as a consequence of Gromov non-squeezing. No such realization is known to the series, and none is asserted.*

9 Assessment

Interpretation 9.1 (Status of the founding thesis after synthesis). *The bounded report the three papers jointly support, and nothing stronger. Precision: for finite-dimensional classical mechanics the thesis is statable, by Definitions 2.1 and 2.2. Internal coherence: the object reproduces the standard formalism (HR thm-local-reconstruction, thm-exact-sequence) without modification. Usefulness: the typing did work; it produced HR’s three new results, CS’s separation equivalence, EL’s error-transfer theorem, and the four cross-paper statements of Section 4, and it converted the methodological preference for interconnection over tensor into a theorem about the range of $\otimes$. Costs, named: $\otimes$ cannot express interaction; the natural richer composition is only partially defined; the structure-preserving arrow class contains no coarse-graining; the type does not extend functorially to quantum theory; and its formal appendix verifies discrete surrogates only. Verdict: we do not report the thesis as established. We report that, in this domain, it is precise, non-trivial, partially falsifiable by (F1)–(F4), and encumbered by five named deficits. Realization remains what CS-II, HR int-locus and EL’s realization criterion each independently make it: empirical input, never a consequence of mathematical admissibility.*

Source status. Bibliographic identity for the items below was verified in the Topic-2 verification record and is inherited here rather than re-verified. Two residuals from that record persist and are repeated so that they are not lost: the end page of Weinstein’s *Symplectic categories* is inferred, not confirmed; and chapter-level pointers into Abraham–Marsden, Arnold, and Marsden–Ratiu were not re-paginated against the physical editions.

References

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